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Riverline: Local-First PWA Storage, Accountless Transfer, Sync Architecture & Privacy Compliance

Extended architectural evaluation integrating the Riverline Project Brief (2026-10-21) for a plain-JavaScript, local-first No-Limit Hold'em study workstation across Desktop Electron and Mobile PWA.

Executive Verdict: How the Riverline Project Brief Reshapes the Research

Integrating `RIVERLINE_BRIEF.md` (dated 2026-10-21) transforms generic local-first research into a domain-specific blueprint tailored to Riverline's exact technical, epistemic, and legal guardrails **[Judgement]**. Without the brief, storage, transfer, and sync are evaluated as generic web plumbing. With the brief, seven concrete architectural constraints govern every design decision:

Brief Constraint	Source Clause in Brief	Architectural Impact on Storage, Transfer & Sync [Judgement]
Plain JS Stack & Node Test Harness	§3: Plain JS (`logic.js` ~14k lines + ES modules), no React/Vue/Tailwind, Node built-in test runner (~2,750 tests), Firefox on Windows checks.	Eliminates React-coupled sync SDKs (ElectricSQL hooks, Triplit React) and WASM-heavy CRDTs (Automerge 320 kB WASM) that complicate bundler-free ES modules and Node test execution.
Closed-Source & Free-First Sourcing	§1 & §2.7: Currently treated as closed-source; zero budget committed; record every outside component's licence before use.	Strict permissive-only filter (MIT, Apache-2.0, BSD, ISC) or zero-dependency Web APIs (`IndexedDB`, `SubtleCrypto`, `CompressionStream`). Bans GPL/AGPL libraries from the client bundle.
Epistemic Honesty & Unknown Preservation	§2.1–2.3: Confidence tiers (`Exact`, `Verified`, `Estimated`, `Unavailable`); `Unknown` is never fold, zero, or excluded.	Prohibits silent field-level CRDT merges on poker hands or ranges that could invent action frequencies or coerce `Unknown` cells into `0%`.
Personal Strategy vs. Observed Play	§2.5: The player's plan (Personal Strategy) is first-class and strictly separate from observed play and baseline.	Requires separate IndexedDB object stores and distinct sync conflict policies: immutable/append-only for played hands vs. versioned snapshots for Personal Strategy trees.
One Canonical Owner for Poker Maths	§2.4: Poker maths live only in `shared/poker-domain/`; the UI formats, never computes.	Sync and storage layers must treat hand records and strategy grids as opaque domain envelopes validated exclusively by `shared/poker-domain/` upon load/import.
Anti-RTA Hard Boundaries	§2.6: Study tool, not real-time assistance; no overlays, no live screen/hand-history reading, no detection evasion.	Favors explicit user-triggered QR/share-sheet/file handoffs over background LAN daemons that could resemble real-time desktop-to-phone assistance bridges.

Brief Constraint	Source Clause in Brief	Architectural Impact on Storage, Transfer & Sync [Judgement]
Offline Solver Packs (M3) & Trilingual UI	§1 & §5: Languages EN, RU, HE (RTL with LTR poker notation); offline `binary/postflop-solver` (AGPL) numerical outputs in M3.	Splits local storage into two tiers: compact user data in IndexedDB (<10 MB, backed up/synced) and large read-only solver packs in OPFS/Cache API (50–500 MB, never synced to user cloud quota).

1. Domain Data Topology: What Riverline Actually Stores and Moves

Before selecting browser storage engines, transfer protocols, or cloud sync schemas, the data model must reflect the five concrete artifact classes defined across Milestones M1, M2, and M3 of `RIVERLINE_BRIEF.md` **[Judgement]**. Unlike collaborative text editors where multiple users type into a single paragraph simultaneously, a No-Limit Texas Hold'em study workstation manages structured, domain-validated records with sharply different mutability, size, and privacy profiles.

Riverline Data Class	Contents & Brief Grounding	Typical Size	Mutability Pattern	Recommended Local Store & Sync Policy [Judgement]
1. Reconstructed Hands & Orbit Sessions	Table rules (6–8 handed 30–60 bb no-rake home games; 9-max 100 bb+ online), seat stacks, hole cards (known or `Unknown`), street action sequences, review tags (§1, §2.2, §6).	350–900 bytes raw JSON per hand (~180–350 bytes compressed); ~35–90 KB per 100-hand session.	Append-only core hand history; mutable user review annotations and tags attached by `handId`.	**IndexedDB** (`hands` & `hand_notes` stores). Sync via QR (1–3 hands), Share Sheet/File (sessions), or Supabase append + LWW notes.
2. Personal Strategy Trees & Ranges	Player's intended strategy per spot (169-combo preflop grids, action mixes, sizing branches, explicit `Unknown` / unanswered spots) kept strictly separate from observed play (§2.2, §2.5, §6 M2).	2–15 KB per spot grid; 100–500 KB for a complete game-type strategy workbook.	Edited deliberately during study sessions; requires atomic spot-level versioning or explicit fork/merge conflict resolution.	**IndexedDB** (`personal_strategies` store). Never silently merge conflicting edits within the same spot matrix; preserve both revisions if concurrent edits occur.
3. Table Rules & Workspace Profiles	Saved game configurations (blinds, antes, straddles, rake cap, buy-in bounds), UI language (`EN`, `RU`, `HE`), and global Motion setting (`Instant`, `Faster`, `Standard`, `Reduced`) (§1, §4, §6 M2).	1–5 KB total JSON across all profiles.	Low-frequency key-value updates.	**IndexedDB** (`profiles` store) with fast boot mirror in `localStorage` for initial theme/motion/locale paint to prevent layout flash.
4. Drills, Orbit Progress & Achievements	Spaced-repetition drill history, question-flow responses, Orbit session logs, and M3 achievement state (§1, §6 M2–M3).	50–500 KB per year of active study.	Append-only event log of completed drill items + derived summary counters.	**IndexedDB** (`drill_events` store). Append-only union merge by `eventId` across devices.

Riverline Data Class	Contents & Brief Grounding	Typical Size	Mutability Pattern	Recommended Local Store & Sync Policy [Judgement]
5. Baseline & Solver Reference Packs (M3)	Precomputed numerical lookup tables from offline runs of `b-inary/postflop-solver` (AGPL) and later `HRC Pro` (§5, §6 M3), scoped with confidence tiers (`Exact`, `Verified`, `Estimated`).	5 MB to 250+ MB of static binary/JSON numerical arrays.	Read-only, immutable reference packages versioned by content hash.	**Origin Private File System (OPFS)** or **Cache API**. Never include in user hand backups or Supabase 500 MB free-tier database sync.

Separating **Mutable User Study Data** (Classes 1–4, stored in IndexedDB, typically 1–10 MB total) from **Read-Only Solver Reference Packs** (Class 5, stored in OPFS or Cache API, potentially 50–250+ MB in M3) is the single highest-leverage architectural decision unlocked by the brief **[Judgement]**. It keeps user export files tiny (under 500 KB for an entire year of hands and Personal Strategy), prevents Supabase's 500 MB free database quota from ever filling up with static solver tables, and allows `shared/poker-domain/` to stream solver arrays from OPFS with near-native file I/O speed (Kanopy Labs Local-First Benchmarks).

2. PWA Installation, Offline Storage Quotas & Eviction in 2026

Because Riverline's phone version is planned as an installable Progressive Web App (PWA) built from the same plain-JS codebase and operating standalone first (§1, §6), its reliability depends on how iOS Safari (WebKit), Android Chrome (Blink), and desktop Firefox/Electron manage origin storage in 2026.

2.1 Installation Mechanics Across iOS Safari and Android Chrome

On Android Chrome and desktop Chromium, when the site serves a valid Web App Manifest (with `display: "standalone"`, icons, and `start_url`) over HTTPS with a registered Service Worker, the browser fires the non-standard `beforeinstallprompt` event on the `Window` object. Riverline can stash this event reference and trigger a calm, non-intrusive install action inside the study menu. By contrast, Safari on iOS and iPadOS does not implement `beforeinstallprompt` (MDN `beforeinstallprompt` compatibility); users must tap the native Safari Share icon and select **Add to Home Screen** (WebKit Storage Policy). In the European Union on iOS 17.4+, alternative browser engines are permitted and apply their own engine storage policies, whereas outside the EU all iOS browsers use WebKit (MDN Storage Quotas).

2.2 Storage Quotas and the 7-Day WebKit ITP Eviction Trap

All modern browsers partition storage per origin (`scheme + hostname + port`) and group IndexedDB, Cache API, Service Worker registrations, and the Origin Private File System (OPFS) under a unified origin quota manager, while keeping synchronous Web Storage (`localStorage` and `sessionStorage`) strictly capped at **5 MiB each (10 MiB combined) per origin** (MDN Storage Quotas). Exceeding either limit throws a synchronous or asynchronous `QuotaExceededError`.

Browser / Runtime	Per-Origin Quota (IndexedDB / OPFS / Cache)	Overall Browser Cap Across Origins	Eviction Triggers in Best-Effort Mode	How `persist()` Is Granted
iOS / iPadOS Safari & Home Screen Web App (Safari 17+)	Up to **~60% of total disk space** for browser tabs and Home Screen Web Apps; **~15%** for embedded non-browser	Up to **80% of total disk** for browser apps; **20%** for non-	1) Overall quota exceeded (LRU whole-origin wipe); 2) OS storage pressure; 3) **Proactive 7-day ITP**	Silent heuristic (no popup); granted primarily when opened as a standalone **Home**

Browser / Runtime	Per-Origin Quota (IndexedDB / OPFS / Cache)	Overall Browser Cap Across Origins	Eviction Triggers in Best-Effort Mode	How `persist()` Is Granted
	`WKWebView` apps (WebKit Policy).	browser apps (WebKit Policy).	**wipe** in regular Safari tabs if no user tap/click occurs for 7 days (MDN).	**Screen Web App** (WebKit Policy).
Android Chrome & Desktop Chromium	Up to **60% of total disk size** in both best-effort and persistent modes (MDN Storage Quotas).	Up to **80% of total disk size** across all origins (MDN Storage Quotas).	Only when combined origins exceed 80% disk cap or device hits severe OS storage pressure (LRU whole-origin deletion). Never time-evicted after 7 days (MDN).	Silent heuristic (no popup); granted automatically if PWA is installed, bookmarked, or has high site engagement (MDN).
Firefox (Windows / Desktop Visual Check Target)	Best-effort: smaller of **10% of disk or 10 GiB** group limit. Persistent: up to **50% of disk (capped at 8 TiB)** (MDN Storage Quotas).	Managed dynamically under profile storage pressure (MDN Storage Quotas).	LRU whole-origin eviction when best-effort origins hit storage pressure (MDN Storage Quotas).	Displays an explicit **UI permission popup** asking the user to allow persistent storage (MDN Storage Quotas).
Electron Host (Desktop Windows / macOS / Linux)	Chromium quota (60% of disk) backed by dedicated OS application data directory (e.g., `%APPDATA%/Riverline`).	Dedicated profile; no shared browser tab competition.	Exempt from browser tab LRU eviction; persists until app uninstallation or explicit user reset.	Can also write native backup snapshots directly to user disk via Electron IPC if desired **[Judgement]**.

Critical Operational Insight for Riverline [Judgement]: When a browser evicts an origin, it deletes *all* data for that origin atomically (both IndexedDB and Cache API together) to prevent inconsistent application state (MDN Storage Quotas). On iOS Safari, if the owner or a friend opens Riverline in a standard browser tab during a weekly home game, closes the tab, and returns 8 days later, Intelligent Tracking Prevention (ITP) can wipe the entire IndexedDB database. Installing Riverline to the iOS Home Screen and calling `navigator.storage.persist()` eliminates this 7-day timer (WebKit Storage Policy). Consistent with Principle #1 (*Honesty is the brand*), Riverline should inspect runtime persistence on boot and surface the exact storage durability state to the user rather than pretending browser tab storage is permanent **[Judgement]**.

2.3 Plain-JS Storage Initialization & Honesty Badge Implementation

Because Riverline uses plain ES modules without external frameworks (§3) and tests in Firefox on Windows (where `navigator.storage.persist()` triggers a visible permission dialog, unlike Chrome/Safari), Riverline should check `navigator.storage.persisted()` passively on boot and invoke `navigator.storage.persist()` either when running standalone or when the user saves their first hand or strategy node **[Judgement]**:

```
// app/src/core/storage-durability.js — Plain ES Module (Zero Dependencies)
export async function probeStorageDurability() {
  const isStandalone = window.matchMedia('(display-mode: standalone)').matches
    || window.navigator.standalone === true
    || navigator.userAgent.includes('Electron');
```

```
  let persisted = false;
  let usageBytes = null;
  let quotaBytes = null;

  if (navigator.storage && navigator.storage.persisted) {
    persisted = await navigator.storage.persisted();
    // Request persistence automatically if already installed as standalone PWA/Electron
    if (!persisted && isStandalone && navigator.storage.persist) {
      persisted = await navigator.storage.persist();
    }
  }
  if (navigator.storage && navigator.storage.estimate) {
    const est = await navigator.storage.estimate();
    usageBytes = typeof est.usage === 'number' ? est.usage : null;
    quotaBytes = typeof est.quota === 'number' ? est.quota : null;
  }

  // Classify durability using Riverline's honest status model [Judgement]
  const isIosTabAtRisk = /iPhone|iPad|iPod/.test(navigator.userAgent) && !isStandalone;
  return {
    persisted,
    isStandalone,
    isIosTabAtRisk, // True = subject to 7-day WebKit ITP script data eviction
    usageBytes,
    quotaBytes
  };
}
```

3. Options for Optional Sync & Local-First Libraries (Filtered by the Brief)

The brief establishes two filters that eliminate most commercial or framework-heavy sync engines **[Judgement]**: (1) Riverline is currently treated as **closed-source** and **free-first** with every external component's licence recorded before use (§1, §2.7), and (2) the codebase is **plain JavaScript** (~14k lines in `logic.js` + ES modules, tested in Node's built-in test runner, with an existing unused Supabase client path) (§3).

3.1 Supabase Free Tier Limits, Pausing Behavior & Row-Level Security (RLS)

Because a Supabase client already exists in the Riverline repository (§3), activating Supabase for optional Phase 3 sync requires zero new client libraries. On the Supabase Free Plan ($0/month), exact 2026 resource limits are:

- **Database & Compute Quota: 500 MB** dedicated Postgres database size per project on shared CPU with 500 MB RAM, unlimited API requests, and a limit of **2 active free projects** (Supabase Pricing). At ~500 bytes per compressed hand record, 500 MB holds roughly **800,000+ reconstructed hands** before hitting the free database ceiling **[Judgement]**.
- **Inactivity Pausing:** Free projects are automatically **paused after 1 week (7 days) of inactivity** and do not include automatic daily backups or Point-in-Time Recovery (which require the $25/month Pro plan with 8 GB disk and 7-day backups) (Supabase Pricing).
- **Auth, Storage & Bandwidth Quotas: 50,000 Monthly Active Users (MAUs)** (including anonymous sign-ins and social OAuth), **1 GB file storage** (max 50 MB per file upload), **5 GB database egress** + 5 GB cached egress, **500,000 Edge Function invocations**, and **Realtime Postgres Changes** capped at 200 concurrent peak connections, 2 million messages/month, and **256 KB max message size** (Supabase Pricing).
- **Row-Level Security (RLS) Requirements:** In Supabase, any table in an exposed schema (such as `public`) without RLS enabled is readable and writable by any client holding the project's public key (Supabase RLS

Docs). You must enable RLS on every table, revoke default `anon` grants if anonymous access is unused, and enforce `auth.uid() = user_id` on `SELECT`, `INSERT`, `UPDATE`, and `DELETE` policies (Supabase RLS Docs). Never embed a Supabase `service_role` secret key in the browser or Electron client, as it bypasses all RLS policies (Supabase API Keys Docs).

3.2 Comprehensive Evaluation of Sync Engines & CRDT Libraries

Evaluating the leading 2026 local-first sync engines and Conflict-free Replicated Data Type (CRDT) libraries against Riverline's plain-JavaScript architecture, closed-source licensing requirement, and zero-budget constraint reveals a sharp divide between lightweight pure-JS primitives and heavy WebAssembly/server-coupled frameworks **[Judgement]**:

Tool / Library	Licence & Closed-Source Legal Status	Free Tier / Hosting Model (2026)	Bundle & Plain-JS / Node Test Compatibility	Riverline Domain Verdict [Judgement]
Supabase (Direct Postgres + LWW / Append Log)	Apache-2.0 / MIT client (`@supabase/supabase-js`). Legal in closed-source apps.	$0/mo Free Tier (500 MB DB, 1 GB storage, 50K MAUs, pauses after 7d inactivity) (Supabase Pricing).	Client already in repo (§3); plain JS/ESM compatible; easy to mock in Node's built-in test runner.	**Recommended for Optional Cloud Sync**: Matches append-only hands + versioned Personal Strategy snapshots with zero new dependencies.
Yjs (Docs)	MIT Licence. Legal in closed-source apps.	100% free open-source library; bring your own transport (WebSockets, WebRTC, IndexedDB, or Supabase bytea blobs).	Pure JavaScript (~18 kB min+gzipped, zero WASM overhead) (PkgPulse 2026); runs cleanly in Node and Firefox.	**Best CRDT if Live Collaborative Study (M3 "Connected Workspaces") Is Added**: Do not use for basic hand storage, as operation logs grow without compaction.
Automerge (Repo)	MIT Licence. Legal in closed-source apps.	100% free library; BYO relay or peer sync via `automerge-repo`.	Rust compiled to WebAssembly (~320 kB gzipped bundle, ~50 ms WASM init, 41 MB RAM on large trace) (PkgPulse 2026).	**Skip**: WASM loading complicates bundler-free plain JS scripts and adds 320 kB weight for Git-like DAG features Riverline does not need.
Loro	MIT Licence. Legal in closed-source apps.	100% free library; manual network and storage wiring required.	Rust + WASM (~180 kB gzipped bundle); fast Fugue CRDT encoding (PkgPulse 2026).	**Skip**: Same WASM packaging friction in a bundler-free plain JS repo and smaller ecosystem (~12K weekly downloads).
ElectricSQL (Pricing)	Apache-2.0. Legal in closed-source apps.	Free self-hosting (Elixir service + Postgres); Electric Cloud PAYG waives bills under **$5/mo** ($1/1M	Streams Postgres "shapes" to client; write-path still requires custom API endpoints (Kanopy Labs).	**Skip for Solo Dev**: Operating an extra Elixir sync tier or configuring shape replication is unnecessary when a

Tool / Library	Licence & Closed-Source Legal Status	Free Tier / Hosting Model (2026)	Bundle & Plain-JS / Node Test Compatibility	Riverline Domain Verdict [Judgement]
		writes, $0.10/GB-mo) (Electric Pricing).		single SQL `updated_at` cursor suffices.
PowerSync (FSL)	Client SDKs Apache-2.0; Open Edition service under FSL-1.1-ALv2 (non-competing use allowed; converts to Apache-2.0 after 2 years).	**$0/mo Cloud Starter**: 500 MB hosted, 2 GB synced/mo, 50 peak clients; **deactivates after 1 week of inactivity** (PowerSync Pricing).	Requires client-side WASM SQLite (~15–20 MB memory footprint, 300–500 ms cold start on web) (Kanopy Labs).	**Skip**: Introduces a second cloud service that also pauses after 7 days of inactivity, plus heavy WASM SQLite initialization on mobile.
Triplit	Open-source monorepo (GitHub; verify package-level licence before use per §2.7).	Self-hosted Node/Durable Objects server (HN Triplit Thread).	TypeScript-centric custom query DSL lacking native SQL `COUNT/UNIQUE` aggregations (HN Thread).	**Skip**: Replaces your existing Supabase setup with a custom server and non-standard query DSL.
Plain Encrypted File Export/Import (Web Crypto + CompressionStream)	W3C / WHATWG built-in browser standards. Zero external licence risk.	100% free forever; zero server infrastructure or recurring cost.	0 kB bundle cost; natively supported in Chrome 80+, Safari 16.4+, Firefox 113+, and Node 18+/21.2+ (MDN).	**Mandatory Foundation (Phase 1 & 2)**: Powers file backup, Share Sheet handoffs, and E2EE payload encryption prior to Supabase upload.

3.3 Why Blind CRDT Merging Violates Riverline's Domain Principles [Judgement]

A critical insight from §2.1–2.5 of the brief is that **poker domain objects have strict structural invariants owned exclusively by `shared/poker-domain/`**. Suppose a player edits a reconstructed hand on their phone (changing Turn action from `CHECK` to `BET 12bb`) and separately edits the same hand on their PC (changing River action assuming a Turn `CHECK`). An automatic attribute-level CRDT merge (such as Triplit, ElectricSQL column LWW, or a naive Y.Map) will combine the phone's Turn `BET 12bb` with the PC's River action, producing an illegal action tree where pot size and remaining effective stacks no longer sum correctly! Similarly, if a Personal Strategy range grid is merged cell-by-cell across two devices without re-validating that action frequencies sum to `100%` (or remain explicitly marked `Unknown` per Principle #2), the merge silently corrupts the player's strategy.

Instead, Riverline should use **Immutable Content-Addressed Revisions** validated by `shared/poker-domain/` **[Judgement]**:

- **Played / Reconstructed Hands:** Treat the core hand action tree as immutable once saved (identified by a deterministic hash or UUIDv7). If a user corrects a reconstructed hand's street actions, write a new `revision_id` pointing to `parent_revision_id`. Review notes and tags live in an attached metadata envelope merged by timestamp.
- **Personal Strategy Nodes:** Version each spot's strategy matrix as a complete, atomic JSON document with a `base_revision_hash` and `updated_at`. During sync or file import, if one device's edit is a direct descendant

of the other's `base_revision_hash`, fast-forward cleanly. If both devices edited the same spot concurrently from the same base hash, **do not silently overwrite or blend cells**—keep both versions and display a calm Riverline resolution prompt: *"Phone and Desktop both updated BTN vs BB 40bb strategy while offline — keep Phone version, Desktop version, or save both as comparison branches."* This directly honors Principle #1 (*Honesty is the brand*) and Principle #5 (*The player's plan is first-class*) **[Judgement]**.

4. Moving Data Phone ↔ PC Without Accounts: Channels, Limits & Plain-JS Implementation

The brief specifies that the phone PWA works standalone first and moves hands by **share sheet, file, or QR** before accounts exist (§1, §6). Below are the exact 2026 technical limits, browser compatibility traps across iOS Safari, Android Chrome, and Firefox/Electron on Windows, and zero-dependency implementation patterns.

Transfer Channel	Hard Technical Limit	Practical Riverline Payload Limit [Judgement]	Cross-Platform Compatibility (iOS / Android / Desktop)	Best Use Case in Riverline [Judgement]
Single Static QR Code (ISO 18004)	Version 40 (177×177 modules, Level L): **2,953 bytes binary** or **4,296 alphanumeric chars** (QRcode.com).	**~800–1,200 bytes** compressed binary (~1,100–1,600 Base45/Base64 chars) = **1 to 4 full NLHE hands** with notes.	Universal display on any screen. Note: `BarcodeDetector` API works on Android Chrome 83+ but is **disabled on iOS Safari 17+ and unsupported on Firefox** (MDN).	Instant single-hand transfer from phone to PC (or friend-to-friend at a live home game night) without internet or cables.
Animated QR Loop (Chained Frames)	2,953 bytes per frame × N frames.	**10–30 KB** over a 5–10 second loop at 4 fps (~800 bytes/frame) = **30–80 hands** or 1 Personal Strategy spot.	Requires in-app camera scanner loop on receiving device (via a permissive MIT/Apache-2.0 JS decoder module).	Optional M2 enhancement for moving a full evening's Orbit session across an air-gapped desk.
OS Share Sheet Outbound (`navigator.share`)	OS transport dependent (AirDrop, Nearby Share, Telegram, email attachments: 25 MB to multi-GB).	**1 KB to 50 MB** `.riverline.json` or encrypted `.riverline` archive (thousands of hands + full Personal Strategy).	`navigator.share({ files })` works on **both iOS Safari 15+ and Android Chrome** (MDN `share()`). Must be triggered by a transient user activation (click/tap).	**Primary Phone → PC / Friend Export**: One tap on phone shares session file over AirDrop, Quick Share, WhatsApp, or Drive.
OS Share Sheet Inbound (`share_target`)	Multipart P0ST intercepted by Service Worker `fetch` handler (MDN `share_target`).	**1 KB to 50 MB** on supported Chromium devices.	Supported on **Android Chrome 76+ and Desktop Chrome/Edge 89+**; **NO support on iOS Safari or Firefox** (MDN `share_target`).	Progressive enhancement for Android users; iOS users import via the in-app File Picker from the iOS Files app.
Direct File Export / Import (`<a download>` & `<input type="file">`)	Browser Blob memory limit (typically 500 MB–2 GB+).	**Unlimited for study data** (10,000 hands + full strategy is <5 MB compressed).	**100% universal** across iOS Safari, Android Chrome, Firefox on Windows, and Electron. Desktop	**Universal Baseline**: Guaranteed fallback and archival backup mechanism on every platform.

Transfer Channel	Hard Technical Limit	Practical Riverline Payload Limit [Judgement]	Cross-Platform Compatibility (iOS / Android / Desktop)	Best Use Case in Riverline [Judgement]
			Chromium/Electron also supports `showOpenFilePicker()`.	
Local Network (WebRTC `RTCDataChannel`)	Default **64 KB** SCTP message size (**256 KB+** in modern browsers; unlimited via chunking) (MDN Data Channels).	**Multi-megabyte** streaming once connected, but requires SDP signaling exchange.	HTTPS PWAs block plain `ws://192.168.x.x` LAN sockets (Mixed Content). Serverless WebRTC requires scanning two QR codes back-and-forth to exchange SDP offers/answers.	**Defer / Skip in M1–M2**: Share Sheet + File + Single QR cover 100% of use cases with a fraction of the UX and code complexity.

4.1 Critical Browser Compatibility Trap: `CompressionStream` and `BarcodeDetector`

To pack multiple No-Limit Hold'em hands into a single QR code or compact file without adding external compression libraries to `app/src/core/logic.js`, the native `CompressionStream` API is ideal—but with a crucial codec caveat verified in MDN's 2026 compatibility tables:

- **Use `"deflate-raw"` (or `"gzip"`), NEVER `"brotli"`:** While Safari 18.4+ and Firefox 147+ added `new CompressionStream("brotli")`, **Chrome Desktop and Chrome Android do not support `"brotli"` in `CompressionStream`** (MDN `CompressionStream` compatibility). By contrast, `"deflate-raw"` (which omits the 6-byte zlib header/checksum wrapper of `"deflate"` and 18-byte wrapper of `"gzip"`) has **full baseline support** across Chrome 103+, Edge 103+, Firefox 113+, Safari 16.4+, Chrome Android 103+, Safari on iOS 16.4+, and Node.js 21.2+ (MDN `CompressionStream`).
- **Handle QR Decoding Without Relying Solely on `BarcodeDetector`:** Because the native `BarcodeDetector` API is unsupported in Firefox on Windows and disabled by default in iOS Safari (MDN `BarcodeDetector`), Riverline should use a two-pronged QR strategy **[Judgement]**:
  1. **Zero-Scanner URL Fragment Handoff (Phone Camera → PWA):** Encode single-hand QR codes as valid HTTPS URLs pointing to the PWA origin with a hash fragment (e.g., `https://riverline.app/#import=v1.<base64url-deflate-payload>`). Because the payload sits after the `#` hash fragment, **the browser never sends the hand data to the server in the HTTP request**, preserving 100% local privacy while allowing users to scan the QR code using their phone's built-in iOS or Android system camera app!
  2. **In-App Camera Scanner Fallback (Phone → PC Webcam):** For scanning a phone screen from a laptop webcam in Firefox/Electron, use feature detection (`if ('BarcodeDetector' in globalThis)`) with a small, permissive MIT/Apache-2.0 pure-JS fallback module recorded in your dependency ledger per Principle #7.

4.2 Bidirectional Language Safety (EN / RU / HE) in Exports & QR Payloads

Because Riverline supports English, Russian, and Hebrew with right-to-left (RTL) prose and left-to-right (LTR) poker notation (§1), player notes frequently mix Hebrew characters (2 bytes per code point in UTF-8), Cyrillic characters (2 bytes in UTF-8), and card suit symbols (♠ ♥ ♦ ♣, 3 bytes each in UTF-8). Two rules prevent subtle corruption across devices **[Judgement]**:

- **Canonical Domain Tokens for Poker Structure:** Store cards as compact ASCII codes (`"As"`, `"Kh"`, `"Td"`, `"2c"`) and actions as enum codes (`"F"`, `"X"`, `"C"`, `"B"`, `"R"`) inside JSON payloads. Render suit glyphs (♠ ♥ ♦

♣) and localized labels (EN/RU/HE) strictly in the UI formatting layer (per Principle #4: *"the UI formats, never computes"*), wrapped in `<span dir="ltr">` when displayed inside a Hebrew `dir="rtl"` view.

- **UTF-8 Byte Streams via `TextEncoder` Before Compression & Encryption:** Never pass raw Unicode strings to `btoa()` (which throws `InvalidCharacterError` on Hebrew, Russian, or suit symbols). Always pipe `new TextEncoder().encode(jsonString.normalize('NFC'))` into `CompressionStream("deflate-raw")` and `SubtleCrypto.encrypt()`.

5. Privacy, Security & Anti-RTA Compliance (GDPR & Israeli Amendment 13)

Because the owner is based in Israel and plans a future release to friends and a global poker community supporting EN, RU, and HE (§1, §6), Riverline falls under the intersection of the European Union's General Data Protection Regulation (GDPR) and Israel's Privacy Protection Law, 5741–1981 (PPL) as overhauled by Amendment No. 13 (which entered into force on August 14, 2025, with active administrative fines issued by the Israeli Privacy Protection Authority [PPA] between July and September 2026) (INPLP Amendment 13 Enforcement Report).

5.1 Legal & Security Comparison: Accountless Phase vs. Cloud Sync Phase

Compliance Dimension	Phase 1 & 2: Standalone PWA + Accountless Transfer (QR / Share Sheet / Encrypted File)	Phase 3: Optional Accounts & Cloud Sync (Supabase Auth + Postgres)
GDPR Status & Scope (Art. 2, Art. 28)	**Out of Controller Scope for User Study Data [Judgement]**: Data never leaves the user's device except via user-directed local transfer. User's own study of home-game hands falls under the purely personal or household activity exemption (Art. 2(2)(c) GDPR). Static web hosting logs only ephemeral IP requests if unlogged.	**Riverline = Data Controller; Supabase = Data Processor**: Storing user emails, auth UUIDs, IPs, and synced records triggers full Controller obligations, requiring a signed Art. 28 Data Processing Agreement (DPA) with Supabase and lawful basis disclosure.
Israeli PPL & Amendment 13 Status (INPLP)	**No "Database of Personal Information" Held by Riverline [Judgement]**: Because zero personal identifiers or study records are collected on a server you control, you are not a "Database Controller" of user hand histories and have no external "Holders" processing user databases.	**Riverline = "Database Controller"; Supabase = External "Holder"**: Amendment 13 broadened "personal information" to include online identifiers (INPLP). You must document Supabase as a Holder in your Database Definitions Document (Reg. 2(a)(7)) and regulate external outsourcing in security procedures (Reg. 15(a)(3)).
Mandatory Transparency Notice (Israel PPL §11 & GDPR Art. 13)	Minimal static-site notice stating that Riverline stores all hands locally on the user's device and collects no personal data on a remote server.	**Strict Statutory Notice Required Before Sign-Up**: In August 2026, the PPA fined a small business NIS 12,000 for an online registration form whose notice omitted controller identity/contact, purposes, recipients, and access/rectification rights under Section 11 (INPLP).
Erasure & Breach Notification Duties (Art. 17, Art. 32)	User controls local storage deletion via a 1-click "Clear Local Workspace" button. Zero server database means zero server data breach risk.	Must provide 1-click account & data erasure (Art. 17 GDPR; NIS 15,000 PPA statutory fine for access/deletion breaches). Serious security incidents require **immediate** PPA notification upon discovery without waiting for internal forensics (July 2026 Meuhedet NIS 256,000 decision; INPLP).

Compliance Dimension	Phase 1 & 2: Standalone PWA + Accountless Transfer (QR / Share Sheet / Encrypted File)	Phase 3: Optional Accounts & Cloud Sync (Supabase Auth + Postgres)
Third-Party Player Notes & Buy-In Privacy Risk [Judgement]	Home-game hand notes often contain friends' names and financial buy-ins (30–60 bb). Encrypting exported files with AES-GCM prevents leaks in cloud drives/chat apps.	If uploaded in plaintext, notes about identifiable home-game players and financial activity could be construed as third-party personal/financial data. **Client-side E2EE before upload** makes synced payloads unreadable ciphertext (Art. 32 GDPR).

5.2 Client-Side End-to-End Encryption (E2EE) Pattern for File Export & Cloud Sync

By using the built-in `SubtleCrypto` interface of the Web Crypto API (`AES-GCM` with a 256-bit key derived via `PBKDF2-SHA256` at 600,000 iterations, a random 16-byte salt, and a random 12-byte IV per encryption), Riverline can use the **exact same encryption function** for both password-protected `.riverline` file exports in Phase 2 and zero-knowledge Supabase ciphertext blobs in Phase 3 **[Judgement]**:

```
// app/src/core/crypto-envelope.js — Plain ES Module using W3C SubtleCrypto + CompressionStream
export async function sealRiverlinePayload(domainObject, passphrase) {
  const jsonBytes = new TextEncoder().encode(JSON.stringify(domainObject).normalize('NFC'));
  const compressedStream = new Blob([jsonBytes]).stream().pipeThrough(new
CompressionStream('deflate-raw'));
  const compressedBytes = new Uint8Array(await new Response(compressedStream).arrayBuffer());

  if (!passphrase) {
    return { v: 1, alg: 'deflate-raw', data: uint8ToBase64Url(compressedBytes) };
  }

  const salt = crypto.getRandomValues(new Uint8Array(16));
  const iv = crypto.getRandomValues(new Uint8Array(12));
  const keyMaterial = await crypto.subtle.importKey(
    'raw', new TextEncoder().encode(passphrase.normalize('NFKC')), 'PBKDF2', false, ['deriveKey']
  );
  const aesKey = await crypto.subtle.deriveKey(
    { name: 'PBKDF2', salt, iterations: 600000, hash: 'SHA-256' },
    keyMaterial,
    { name: 'AES-GCM', length: 256 },
    false,
    ['encrypt']
  );
  const ciphertext = new Uint8Array(await crypto.subtle.encrypt({ name: 'AES-GCM', iv }, aesKey,
compressedBytes));
  return {
    v: 1,
    alg: 'deflate-raw+aes-256-gcm+pbkdf2-sha256',
    iter: 600000,
    salt: uint8ToBase64Url(salt),
    iv: uint8ToBase64Url(iv),
    ct: uint8ToBase64Url(ciphertext)
  };
}
```

6. Recommended Phased Plan for Riverline (Aligned to M1 → M2 → M3)

For a one-developer project maintaining ~14,000 lines of plain JavaScript and ~2,750 Node tests while preparing M1 (trustworthy polished base), M2 (Personal Strategy, Hand Reconstruction, Review, Game Night), and M3 (Solver Library, Orbit Sessions, Connected Workspaces), the following four-stage plan maximizes velocity and keeps recurring infrastructure costs at **$0/month [Judgement]**.

Stage & Milestone	Concrete Engineering Deliverables	Storage & Schema Architecture	Verification & Principle Alignment [Judgement]
Stage 1: Local Durability & Dual-Tier Storage (M1 / Early M2)	Add Web App Manifest + offline Service Worker; call `probeStorageDurability()` on boot; show honest iOS Safari tab warning banner if `!isStandalone`; create separate IndexedDB stores for `hands`, `personal_strategies`, `profiles`, and `drill_events`.	Every record carries `id` (UUIDv7), `schema_version`, `revision_hash`, `parent_revision_hash`, `updated_at`, and `deleted_at` (tombstone). Keep fast UI boot settings (locale `EN/RU/HE`, Motion setting) mirrored in `localStorage`.	Honors Principle #1 (*Honesty is the brand*—never hide iOS 7-day tab eviction risk) and Principle #5 (strictly separate stores for Personal Strategy vs. Observed Play). Testable in Node via an in-memory IndexedDB adapter.
Stage 2: Accountless Transfer Suite (M2 Game Night & Review)	Build Single-Hand QR (compress canonical hand JSON via `CompressionStream("deflate-raw")` into URL hash `#import=v1...` under 800 bytes) and Share Sheet / File Export-Import via `navigator.share({ files })` and `<input type="file">` with optional AES-GCM passphrase.	On import, validate every hand and Personal Strategy node through `shared/poker-domain/` before writing to IndexedDB. Deduplicate hands by `id`; if a Personal Strategy spot has divergent concurrent edits, retain both branches for user choice.	Honors Principle #2 (`Unknown` cells preserved bit-for-bit), Principle #4 (domain validates all imported maths), Principle #6 (explicit post-hand transfer, zero RTA behavior), and Principle #7 (zero paid dependencies).
Stage 3: OPFS Solver Pack Cache (M3 Expansion)	Store offline-generated numerical lookup packs from `b-inary/postflop-solver` (AGPL) in the Origin Private File System (OPFS) via `navigator.storage.getDirectory()`.	Content-addressed binary/JSON files in OPFS (e.g., `/solver-packs/srp-btn-bb-40bb-v1.bin`). Excluded from user hand exports and cloud sync.	Keeps user backups under ~500 KB even when M3 solver libraries reach 100+ MB; legally separates AGPL offline solver generation from the closed-source app (§5).
Stage 4: Optional E2EE Supabase Sync (Post-M2 / M3 Side Track)	Enable existing Supabase client behind an opt-in "Cloud Backup & Sync" toggle; encrypt rows client-side with user's Sync Passphrase via `sealRiverlinePayload()`; sync deltas via `updated_at > last_cursor`.	Single Postgres table `sync_envelopes (user_id UUID, entity_type TEXT, entity_id UUID, revision_hash TEXT, parent_hash TEXT, updated_at TIMESTAMPTZ, deleted_at TIMESTAMPTZ, sealed_blob TEXT)` with strict RLS (`auth.uid() = user_id`).	Zero server knowledge of poker hands, opponent names, or buy-ins; fits 800,000+ hands in Supabase's 500 MB free tier; includes mandatory Section 11 / GDPR privacy notice and 1-click account deletion RPC.

7. Risk Register, Mitigations & Open Questions for the Owner

Following §7 of the brief (*"Keep outputs structured: a short answer first, then the evidence table, then risks and open questions"*), the table below consolidates the technical, legal, and product risks alongside concrete mitigations and open decisions for the owner.

Risk Category	Failure Mode & Evidence	Severity / Likelihood	Concrete Mitigation [Judgement]
1. iOS Safari 7-Day ITP Data Wipe	User opens Riverline in a standard iOS Safari tab (not installed to Home Screen), logs home-game hands, and does not open the tab for 7 days; WebKit proactively deletes all script-created IndexedDB/OPFS storage (MDN Storage Quotas).	**High Severity / High Likelihood**	Detect `iOS && !window.navigator.standalone` on boot; display a prominent calm banner guiding the user to tap **Share → Add to Home Screen**, and show an honest indicator (`Storage: Temporary Browser Tab · Back up or install`) until `navigator.storage.persist()` resolves to `true`.
2. Silent Strategy or Hand Corruption on Merge	Two devices edit the same Personal Strategy spot or hand reconstruction offline; a field-level merge blends action frequencies so they no longer sum to 100% or overwrites `Unknown` cells with `0`, violating §2.1–2.5.	**High Severity / Medium Likelihood**	Never merge inside a poker hand or range matrix at the field level. Use content-addressed revision hashes (`revision_hash` + `parent_revision_hash`), validate every payload through `shared/poker-domain/`, and surface concurrent branches side-by-side for user selection.
3. Cross-Browser API Gaps (`brotli`, `BarcodeDetector`, `share_target`)	1) `CompressionStream("brotli")` fails on Chrome/Android (MDN); 2) `BarcodeDetector` fails on iOS Safari and Firefox on Windows (MDN); 3) Manifest `share_target` fails on iOS Safari (MDN).	**Medium Severity / Certain if Unchecked**	1) Standardize strictly on `CompressionStream("deflate-raw")`; 2) Format single-hand QR codes as `#import=...` URLs scannable by the native iOS/Android camera app, plus a tiny MIT/Apache-2.0 JS scanner fallback for PC webcams; 3) Always provide an explicit in-app **Import File** button alongside `share_target`.
4. Supabase Free Tier 7-Day Inactivity Pause	During early testing with only the owner and close friends, 7 days without a cloud sync request causes Supabase to pause the free Postgres instance (Supabase Pricing).	**Low–Medium Severity / High Likelihood**	Design cloud sync as an asynchronous background queue that fails gracefully when offline or paused (never blocking local study), or schedule a lightweight weekly health-check ping if active testing is underway.
5. Israeli Amendment 13 & GDPR Exposure Once Accounts Launch	Launching Supabase Auth without a complete Section 11 notice, Holder documentation (Reg. 2(a)(7) & Reg. 15(a)(3)), or RLS exposes the owner to statutory PPA administrative sanctions (INPLP 2026 Decisions).	**High Severity / Low Likelihood (if mitigated)**	Keep Phase 1 & 2 100% accountless. Before turning on Phase 3 accounts, ship client-side AES-GCM encryption (`sealRiverlinePayload`), enable Postgres RLS on all tables, publish a Section 11 + GDPR Art. 13 privacy notice in EN/RU/HE, document Supabase as a Holder, and wire a 1-click account deletion endpoint.

Open Questions for the Owner [Judgement]

- **Q1 — QR Scanner Dependency Approval (§2.7 Sourcing Policy):** Generating a QR code in plain JS and scanning it via the phone's native OS camera (using a `#import=` URL hash) requires only a tiny (~8 kB) zero-dependency QR encoder module (MIT). Do you also want PC-webcam scanning inside Firefox/Electron (which requires recording a ~40 kB Apache-2.0/MIT JS QR decoder module in the licence ledger), or is Phone → PC transfer primarily via Share Sheet / File while QR is PC → Phone and Phone → Phone?
- **Q2 — Personal Strategy Conflict UX (§2.5):** When importing a `.riverline` file (or syncing in Phase 3) where a Personal Strategy spot was modified on both devices since their last common ancestor, should

Riverline default to keeping the incoming version as a named comparison draft (e.g., *"BTN vs BB (Imported 2026-10-23)"*) so zero work is ever lost and zero modal interruption blocks the import?

- **Q3 — M3 Solver Pack Distribution (§5):** When shipping offline `b-inary/postflop-solver` numerical packs in M3, will users download optional spot packs on demand into OPFS, or bundle a core baseline pack directly inside the initial PWA/Electron asset cache?

8. Primary Sources & Verification References

- **WebKit Storage Policy & Home Screen Web Apps:** Apple WebKit Team, Updates to Storage Policy (Safari 17+ Origin & Overall Quotas, `StorageManager.persist()`).
- **Browser Storage Quotas, LRU & 7-Day ITP Eviction:** Mozilla Developer Network, Storage quotas and eviction criteria — Web APIs | MDN (updated January 2026) and StorageManager: `persist()` method — MDN (updated October 2026).
- **Web Share, Share Target, Compression, Crypto, File System & WebRTC Standards:** MDN Web Docs, Navigator: `share()` method, Web App Manifest: `share_target` compatibility, `CompressionStream` compatibility (`deflate-raw` vs `brotli`), `BarcodeDetector` compatibility, `SubtleCrypto.encrypt()` (AES-GCM), `Window.showOpenFilePicker()`, and Using WebRTC data channels (message size limits).
- **QR Code ISO/IEC 18004 Capacity Specifications:** DENSO WAVE, Information capacity and versions of QR Code (Version 1 to Version 40).
- **Supabase Free Tier & Security Documentation:** Supabase Inc., Supabase Pricing & Plan Comparison (2026), Postgres Row Level Security (RLS) Guide, and API Keys Security Guide.
- **CRDT & Local-First Sync Benchmarks, Licences & Pricing:** PkgPulse, Yjs vs Automerge vs Loro: CRDT Libraries 2026; Kanopy Labs, Electric SQL vs PowerSync vs LiveStore: Local-First Sync 2026; Yjs, Yjs MIT Licence; Automerge, Automerge MIT Licence; ElectricSQL, Electric Cloud Pricing; PowerSync, PowerSync Pricing and PowerSync Functional Source License (FSL-1.1-ALv2); Triplit, aspen-cloud/triplit Monorepo.
- **GDPR & Israeli Privacy Protection Law (Amendment 13):** Official Journal of the EU, Art. 2 GDPR (Material Scope), Art. 17 GDPR (Right to Erasure), Art. 28 GDPR (Processor), and Art. 32 GDPR (Security of Processing); International Network of Privacy Law Professionals (INPLP), Israel's First Administrative Fines under Amendment 13: A New Enforcement Era for Privacy Compliance (October 8, 2026); TrustIZ, English Translation of the Israeli Privacy Protection Law, 5741–1981 (as amended by Amendment No. 13).